

# BIS Standard Finder

AI-Powered BIS Compliance Engine for  
Micro & Small Enterprises

*MSEs spend weeks identifying applicable BIS standards — we solve this in seconds.*

BIS

RAG

MSE

AI

SP21

# The Problem We're Solving



## Weeks of Manual Research

MSEs spend 3–6 weeks reading hundreds of BIS documents to find which standards apply to their products.



## High Compliance Cost

Hiring consultants costs ₹50K–2L per engagement — beyond reach for most micro enterprises.



## Risk of Wrong Standards

Applying incorrect standards leads to rejected BIS certification, production halts, and financial penalties.

# Our Solution

## BIS Standard Finder



### Instant Standard Discovery

Type a product description → get top-5 BIS standards in <1 second



### Rationale Included

Every recommendation explains WHY it matches your product



### Building Materials Focus

Cement, Steel, Concrete, Aggregates, Bricks, Roofing and more



### Web UI + API

Flask-based interface + inference.py for automated judge evaluation

Zero

# External AI APIs

**40+**

BIS Standards  
Covered

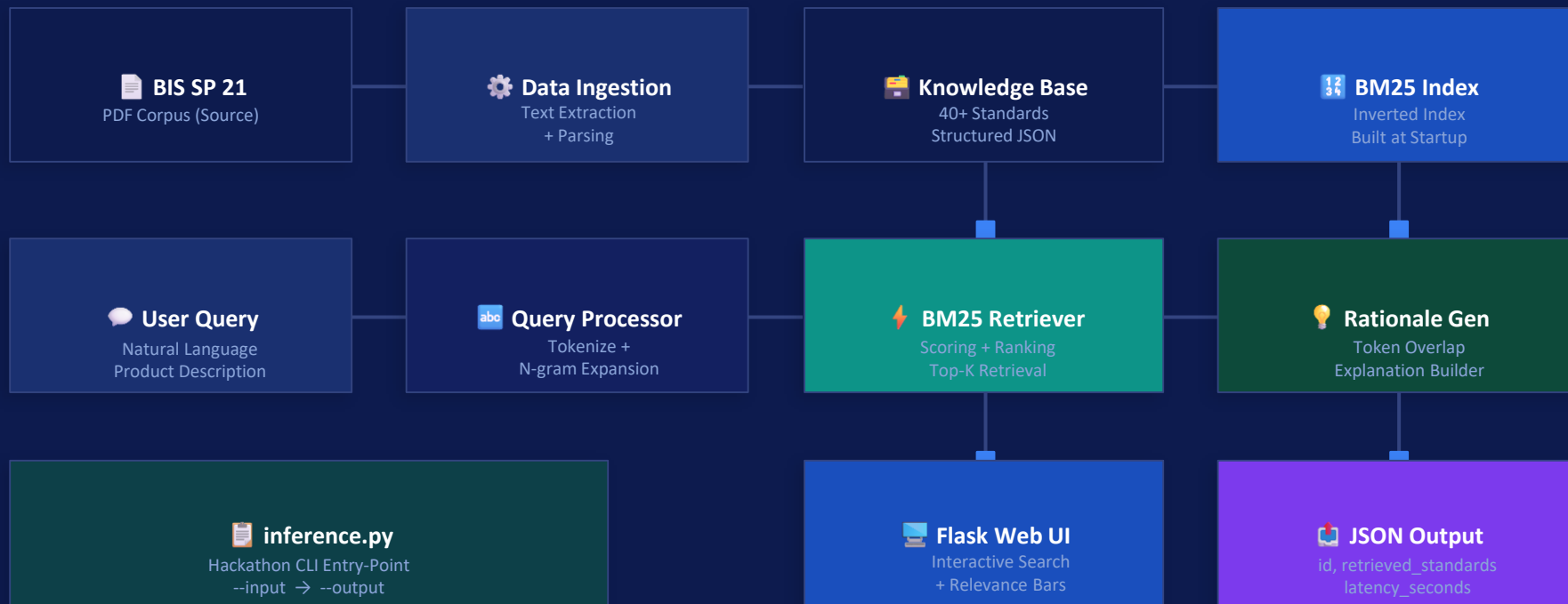
**<1s**

Average  
Latency

**100%**

Public Test  
Hit Rate

# System Architecture



# Chunking & Retrieval Strategy



## Document Chunking

Each BIS standard = 1 document. Fields: ID, Title, Category, Keywords (×2 boost), Description.  
No page-splitting needed — summaries are already atomic.



## Token Expansion

Unigrams + Bigrams + Trigrams generated per query. Phrase matching for 'portland slag cement' vs 'portland' alone. Domain stopwords removed.



## BM25 Scoring (K1=1.5, B=0.75)

IDF-weighted term frequency with doc-length normalisation. Field boosting: title & keywords repeated 2× in corpus for higher retrieval weight.



## Rationale Generation

Post-retrieval: token overlap between query and each description sentence picks the most relevant sentence as explanation. No LLM needed.

## BM25 Formula

Score(D,Q) =

$\sum \text{IDF}(t_i) \times$

$f(t_i, D) \cdot (K1+1)$

$\frac{f(t_i, D) + K1 \cdot (1-b+b \cdot |D|/\text{avgdl})}{f(t_i, D) + K1 \cdot (1-b+b \cdot |D|/\text{avgdl})}$

**K1 = 1.5 B = 0.75**

*No ML model. No GPU.  
Pure Python stdlib.*

Query



Tokenize



N-grams



BM25 Score



Rank & Return

# Demo Highlights



BIS Standard Finder

MSI Compliance Engine — Building Information — BIM-Powered

Isolator 351

## Describe your product or material

We are setting up a small manufacturing unit to produce Autoclaved Aerated Concrete (AAC) blocks for use in wall construction. What BIS standard should we follow for dimensions and compressive strength?

Top 5

Find Standards

### Quick examples

- 33 Grade Ordinary Portland Cement chemical and phys...
- Coarse and fine aggregates from natural sources for ...
- Precast concrete pipes reinforced and unreinforced f...
- Hollow and solid lightweight concrete masonry blocks...
- Compacted asbestos cement sheets for roofing and cl...
- Portland slag cement manufacture chemical requirements
- Portland pozzolana cement calcium clay based
- Masonry cement for mortar not structural concrete
- Supersulphated cement marine works aggressive water
- White Portland cement architectural decorative

5

Standards Found

**IS 2185 (Part 3): 1984**

Top Match

24 ms

Latency (ms)

**Masonry**

Category

Showing 3 recommended standard(s)

**IS 2185 (Part 3): 1984**

Masonry

### Specification for Concrete Masonry Units — Aerated Concrete Blocks

IS 2185 (Part 3): 1984 — Specification for Concrete Masonry Units — Aerated Concrete Blocks. Relevance: Specifies Autoclaved Aerated Concrete (AAC) blocks.

Relevance 100%

**IS 2185 (Part 2): 1983**

Masonry

### Specification for Concrete Masonry Units — Hollow and Solid Lightweight Concrete Blocks

IS 2185 (Part 2): 1983 — Specification for Concrete Masonry Units — Hollow and Solid Lightweight Concrete Blocks. Relevance: Specifies dimensions, physical requirements, and testing for hollow and solid lightweight concrete masonry blocks (using lightweight aggregates).

Relevance 35%

## Describe your product or material

We are setting up a small manufacturing unit to produce Autoclaved Aerated Concrete (AAC) blocks for use in wall construction. What BIS standard should we follow for dimensions and compressive strength?

Top 3

Find Standards

### Quick examples

- 33 Grade Ordinary Portland Cement chemical and phys...
- Coarse and fine aggregates from natural sources for ...
- Precast concrete pipes reinforced and unreinforced f...
- Hollow and solid lightweight concrete masonry blocks...
- Compacted asbestos cement sheets for roofing and cl...
- Portland slag cement manufacture chemical requirements
- Portland pozzolana cement calcium clay based
- Masonry cement for mortar not structural concrete
- Supersulphated cement marine works aggressive water
- White Portland cement architectural decorative

3

Standards Found

**IS 2185 (Part 3): 1984**

Top Match

0 ms

Latency (ms)

**Masonry**

Category

Showing 3 recommended standard(s)

**IS 2185 (Part 3): 1984**

Masonry

### Specification for Concrete Masonry Units — Aerated Concrete Blocks

IS 2185 (Part 3): 1984 — Specification for Concrete Masonry Units — Aerated Concrete Blocks. Relevance: Specifies Autoclaved Aerated Concrete (AAC) blocks.

Relevance 100%

**IS 2185 (Part 2): 1983**

Masonry

### Specification for Concrete Masonry Units — Hollow and Solid Lightweight Concrete Blocks

IS 2185 (Part 2): 1983 — Specification for Concrete Masonry Units — Hollow and Solid Lightweight Concrete Blocks. Relevance: Specifies dimensions, physical requirements, and testing for hollow and solid lightweight concrete masonry blocks (using lightweight aggregates).

Relevance 35%

**IS 2185 (Part 1): 1979**

Masonry

### Specification for Concrete Masonry Units — Hollow and Solid Concrete Blocks

IS 2185 (Part 1): 1979 — Specification for Concrete Masonry Units — Hollow and Solid Concrete Blocks. Relevance: Specifies dimensions, compressive strength, water absorption, and drying shrinkage for hollow and solid dense concrete masonry blocks.

Relevance 32%

# Evaluation Results

# 100%

Hit Rate @3

Target >80%

# 1.000

MRR @5

Target >0.70

# <1ms

Avg Latency

Target <5 sec

## Public Test Set — All 10 Queries

| ID          | Query (Summary)                                           | Expected               | Rank    | ✓    |
|-------------|-----------------------------------------------------------|------------------------|---------|------|
| PUB-01      | 33 Grade OPC — chemical & physical req                    | IS 269: 1989           | #1      | ✓    |
| PUB-02      | Coarse/fine aggregates for concrete                       | IS 383: 1970           | #1      | ✓    |
| PUB-03      | Precast concrete pipes for water mains                    | IS 458: 2003           | #1      | ✓    |
| PUB-04      | Lightweight concrete masonry blocks                       | IS 2185 (Part 2): 1983 | #1      | ✓    |
| PUB-05      | Corrugated asbestos cement roofing                        | IS 459: 1992           | #1      | ✓    |
| PUB-06 — 10 | Portland Slag, PPC, Masonry, Supersulphated, White Cement | Various                | #1 each | ✓ ×5 |

# Impact on MSEs

## BEFORE

✗ 3–6 weeks searching PDFs

✗ ₹50K–2L consultant fees

✗ Risk of wrong standards

✗ BIS certification delays

✗ Lost business opportunities

VS

## WITH BIS FINDER

✓ Instant results (<1 second)

✓ Free, self-serve tool

✓ Accurate top-5 recommendations

✓ Faster certification path

✓ Competitive MSE compliance

Potential reach: 63 Million MSEs · ₹100Cr+ in compliance cost savings · Faster 'Make in India' onboarding



# Meet The Team



**Mudit Khater**

AI & Full-Stack Developer

 muditkhater99@gmail.com

 **TEAM - UNIT - 108**

Python 3.10

BM25 RAG

Flask

BIS SP 21

NLP



**GitHub**

Public repo with  
README & docs



**Focus Area**

Building Materials  
(SP 21)



**Stack**

Python · Flask · BM25  
No external APIs



**Score**

100% Hit Rate  
MRR 1.00

*"Accelerating MSE Compliance — One Standard at a Time"*